

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	B.Jahnavi	Reg. No.:	192325043
Section / Batch:	4 th year /AIML	Date:	09/08/26

Code of Conduct

I B.Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.Jahnavi

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Objective

A search engine should group semantically related words under a unified representation. The morphological analyzer identifies the root, prefixes/suffixes, morphological type, and normalized form.

Morphological Rules

-ing → indicates an inflectional form such as a present participle.

-sis in analysis is part of the derived noun form and is not simply removed as an English suffix.

-ical → derivational suffix that forms an adjective.

The common lexical root/base is analyze/analyse conceptually, with spelling variation between analyze and analysis.

Analysis

Word	prefix	Root	suffix	Type	Normalized representation
analyzing	--	analyze	-ing	inflectional	analyze
analysis	--	analy / analysis	--sis	Derivational / lexical	analyze
analytical	--	analysis	-ical	Derivational	analyze

Detailed Breakdown

1. analyzing

analyze + ing → analyzing

Root/base: analyze

Suffix: -ing

Type: Inflectional

Meaning remains related to the verb analyze.

Normalized form: analyze

2. analysis

analyze → analysis

Noun form related to the verb analyze.

It is an irregular/lexically derived nominal form rather than a simple suffix-stripping case.

Type: Derivational

Normalized representation: analyze

3. analytical

analysis + ical → analytical

Base: analysis

Suffix: -ical

Type: Derivational

Changes the word into an adjective.

Normalized representation: analyze

Final Output

analyzing → analyze

analysis → analyze

analytical → analyze

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

Objective

The sentiment analysis system must identify prefixes, suffixes, and base forms while understanding how derivation changes meaning.

Morphological Rules

- dis- + verb → reverses or negates the meaning.
- verb + -ment → forms a noun expressing a state, result, or process.
- verb + -able → forms an adjective expressing possibility or suitability.

Analysis

Word	Prefix	Root	Suffix	Type	Semmentic effect
disagree	dis--	agree	--	Derivational	Reverses/negates agreement
agreement	--	agree	--ment	Derivational	Forms a noun/state
agreeable	--	agree	--able	Derivational	Forms an adjective

Detailed Breakdown

1. disagree

- dis- + agree → disagree
- Prefix: dis-
- Root: agree
- Suffix: none
- Type: Derivational
- Semantic effect: changes positive agreement into disagreement/negation.

2. agreement

- agree + ment → agreement
- Root: agree
- Suffix: -ment
- Type: Derivational
- Changes the verb into a noun referring to the state or result of agreeing.

3. agreeable

- agree + able → agreeable
- Root: agree
- Suffix: -able
- Type: Derivational
- Changes the verb into an adjective meaning acceptable or pleasing.

Final Output

disagree → dis- + agree → agree

agreement → agree + -ment → agree

agreeable → agree + -able → agree

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

Objective

The system must standardize related words for topic modeling, clustering, and lexical normalization.

Morphological Rules

govern → base verb.

govern + -ment → noun referring to the system or process of governing.

govern + -ance → noun referring to the process or manner of governing.

Analysis

Word	Root	Suffix	Type	Word class	Normalized form
govern	govern	---	Base form	Noun	govern
government	govern	---ment	Derivational	Noun	govern
governance	govern	---ance	Derivational	Noun	govern

Detailed Breakdown

1. govern

govern

Base/root: govern

No affix.

It is the basic verb form.

Normalized form: govern

2. government

govern + ment → government

Root: govern

Suffix: -ment

Type: Derivational

Changes the verb into a noun.

Normalized form: govern

3. governance

govern + ance → governance

Root: govern

Suffix: -ance

Type: Derivational

Produces a noun referring to the process or manner of governing.

Normalized form: govern

Final Output

govern → govern

government → govern + -ment → govern

governance → govern + -ance → govern

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

Objective

The system must identify derivational layers, word-class changes, and semantic shifts, and then map related forms to a unified representation.

Morphological Rules

active + -ate → activate: creates a verb.

activate + -ion → activation: creates a noun.

re- + activation → reactivation: adds the meaning “again”.

Prefixes and suffixes are derivational because they create new lexical forms or change meaning/word class.

Analysis

word	Morphological breakdown	Derivational layer	Word class	Meaningful Effect	Normal root
activate	active+-ate	-ate	Verb	Causes something to become active	active
activation	Activate+-ion	-ate+-ion	Noun	Process/state of activating	active
reactivation	Re-+activation	re - + -ate+-ion	Noun	Activation again	active

Detailed Breakdown

1. activate

active + ate → activate

Base: active

Suffix: -ate

Type: Derivational

Changes an adjective into a verb.

Normalized representation: active

2. activation

activate + ion → activation

Base: activate

Suffix: -ion

Type: Derivational

Changes the verb into a noun.

Normalized representation: active

3. reactivation

re + activation → reactivation

Expanded:

re- + active + -ate + -ion → reactivation

Prefix: re-
Root/base: active
Suffixes: -ate, -ion
Type: Derivational
re- adds the meaning “again”.
The word is a noun.
Normalized representation: active

Final Output

activate → active + -ate → active

activation → active + -ate + -ion → active

reactivation → re- + active + -ate + -ion → active

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect. Identify suffix patterns and map all variations to a common base form without altering meaning. Apply the process to extract root forms and classify each transformation. Determine the morphological breakdown and normalized output for each word.

Objective

The search optimization system must handle grammatical variations without changing the underlying meaning. The common base form is create.

Morphological Rules

create → base verb.
create + s → creates → third-person singular present.
create + ing → creating → present participle/continuous aspect.
These suffixes are inflectional because they express grammatical information without creating a fundamentally new lexical meaning.

Analysis

word	root	suffix	Grammatical function	Type	Normalized form
create	create	--	Base verb	Base form	create
creates	create	-s	Third-person singular present	Inflectional	create
creating	create	-ing	Present participle/continuous aspect	Inflectional	create

Detailed Breakdown

1. create

create

Root/base: create
No suffix.
It is the base form.
Normalized form: create

2. creates

create + s → creates

Root/base: create

Suffix: -s

Indicates third-person singular present tense.

Type: Inflectional

Normalized form: create

3. creating

create + ing → creating

Root/base: create

Suffix: -ing

Indicates present participle/continuous aspect.

Type: Inflectional

Normalized form: create

Final Output

create → create

creates → create + -s → create

creating → create + -ing → create

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____